higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRICAL SYSTEMS AND CONSTRUCTION
NQF LEVEL 2

22 NOVEMBER 2019

This marking guideline consists of 6 pages.

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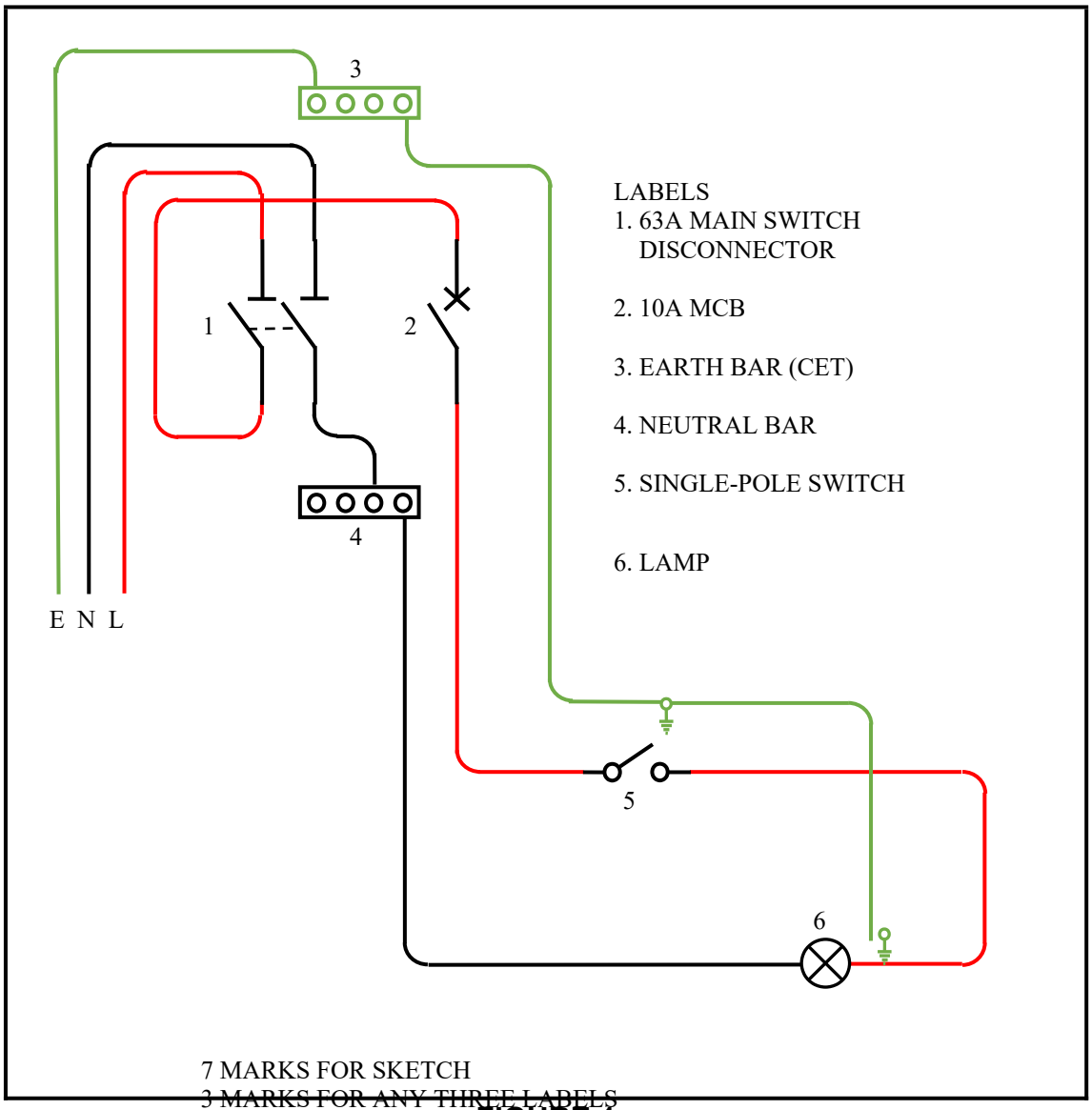
QUESTION 1

1.1

1.1.1	B
1.1.2	C
1.1.3	D
1.1.4	D
1.1.5	C
1.1.6	D
1.1.7	D
1.1.8	A
1.1.9	B
1.1.10	C

(10 × 1) (10)

1.2



7 MARKS FOR SKETCH
3 MARKS FOR ANY THREE LABELS

FIGURE 1 (10)

1.3
- Tools and materials left lying around
- Overcrowded passageways
- Excessive materials or waste in work area
- Dirty and wet work surfaces
- Overflowing waste containers
- Untidy or dangerous storage of materials
- Cluttered and poorly arranged work areas
- Dangerous storage of items overhead
- Items stored on ladders or stairs
- Storage of obsolete items (Any 5 × 1) (5)

[25]

QUESTION 2

2.1 2.1.1 Steel wire-armoured cable. (1)

2.1.2 A – PVC outer sheath
B – Wire armouring
C – Conductor (3)

2.1.3 The wire armouring protects the cable from mechanical damage✓ and serves as a continuous earth.✓ (2)

2.1 2.2.1 Scotch cast joint (1)

2.2.2
- Make off the cable ends.
- Cut the conductors to required length.
- Strip insulation, join conductors using ferrules and reinsulate joints.
- Fit the mould over the joints.
- Tape the ends of the mould to ensure resin does not leak out.
- Mix and pour resin into mould. (6)

2.3 No.✓
A continuous conductor should have a low ohm reading.✓ (2)

2.4
- It is neat
- Cost-effective
- Easy to install
- Conductors can be laid in
- Easy to add or change wiring
- Provides easy access to wiring
- No need to pull conductors through (Any 5 × 1) (5)

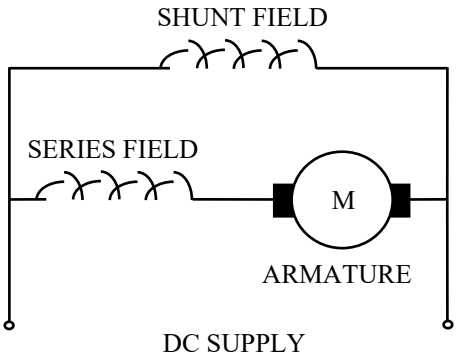
[20]

QUESTION 3

3.1
- Continuity test.
- Insulation resistance test between windings.
- Insulation resistance test between windings/components and earth. (3)

3.2
- Running winding.
- Starting winding.
- Centrifugal switch.
- Running capacitor.
- Starting capacitor. (Any 4 × 1) (4)

3.3



3 MARKS FOR CORRECT CONNECTION OF COILS AND ARMATURE
2 MARKS FOR ANY TWO LABELS

FIGURE 2 (5)

3.4
- Starting
- Stopping
- Reversing
- Protection against damage
- Speed control
- Running
- Safety of operator (Any 3 × 1) (3)

[15]

QUESTION 4

4.1 4.1.1 Earth leakage relay tester
4.1.2 25–30 mA
4.1.3 Multimeter/ohmmeter
4.1.4 0 Ω/low ohm reading
4.1.5 Insulation resistance tester/megger
4.1.6 Not less than 1 MΩ
4.1.7 Multimeter/ohmmeter
4.1.8 0 Ω/low ohm reading
4.1.9 Insulation resistance tester/megger
4.1.10 Not less than 1 MΩ

(10 × 1) (10)

4.2 To ensure that the live conductor✓ is connected to the centre terminal✓ of an Edison screw lamp holder.✓ (3)

4.3 No.✓
An infinity reading should be recorded in the off position.✓ (2)
[15]

QUESTION 5

5.1 5.1.1 True
5.1.2 False
5.1.3 False
5.1.4 False
5.1.5 True
5.1.6 False
5.1.7 False
5.1.8 True
5.1.9 False
5.1.10 True

(10 × 1) (10)

5.2

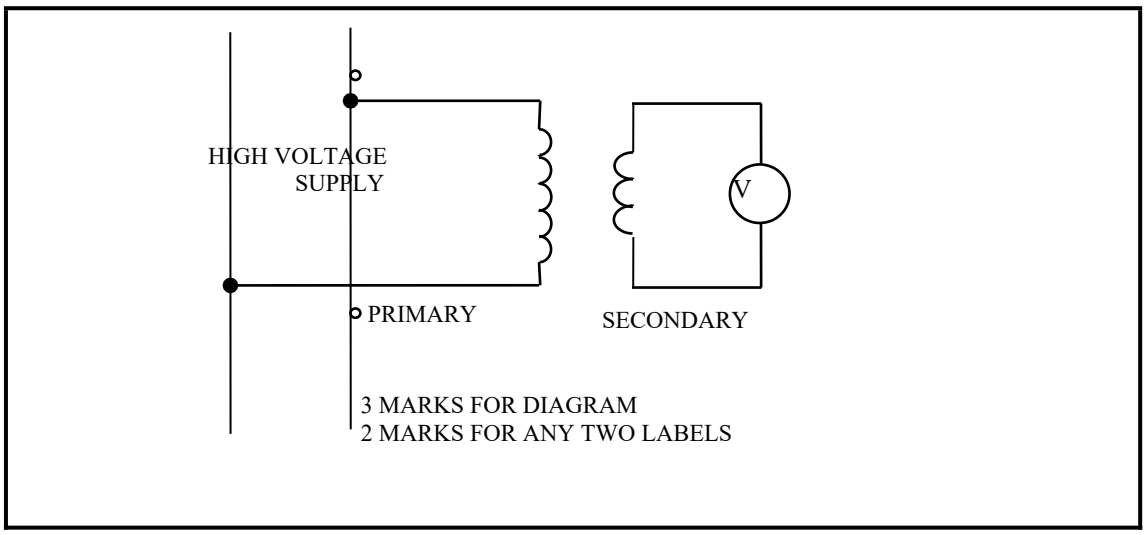


FIGURE 3 (5)

[15]

QUESTION 6

6.1 6.1.1

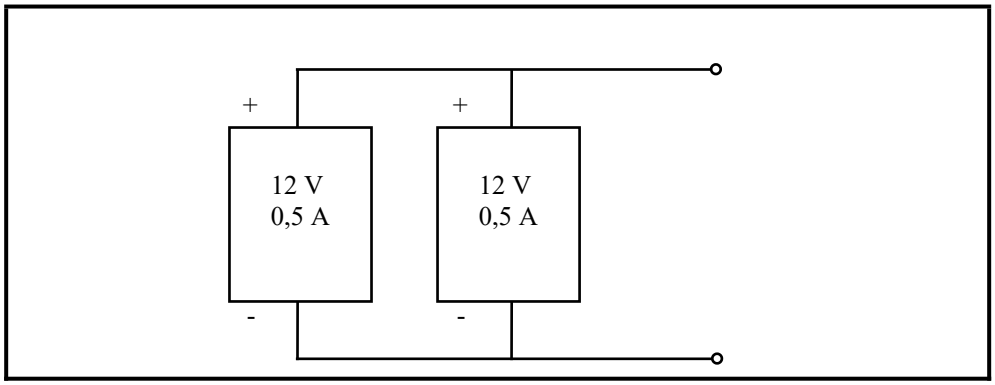


FIGURE 4 (2)

6.1.2 $E_T = E_1 = E_2 = 12$ V (1)

6.1.3 $I_T = I_1 + I_2$
$I_T = 0,5 + 0,5$✓
$I_T = 1$ A✓ (2)

6.2 6.2.1 shines
6.2.2 photons
6.2.3 negatively
6.2.4 free
6.2.5 electricity

(5 × 1) (5)

[10]

TOTAL: 100